William Kwako

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EDUCATION

M.S., Computer Science – Interactive Intelligence, Georgia Tech, GPA: 4.0 August 2023 – August 2026
B.A., Physics, Willamette University August 2015 – May 2019

PROFESSIONAL EXPERIENCE

Technical Solutions Engineer – Epic Systems Corporation December 2019 – December 2022

- Served as point of contact for all claims-related issues across five health system clients; led weekly and biweekly meetings with revenue cycle analysts and managers to triage and resolve issues
- Deployed, maintained, and debugged revenue cycle software, including HIPAA-mandated EDI claims and remittance transactions (837I/837P, 835)
- Coordinated large-scale claims implementations (claim attachments, PLBs, rapid reconciliation, state reporting) to meet standardized claim conformance requirements
- Wrote M/MUMPS queries against production healthcare databases containing PHI to diagnose client issues and analyze data for resolution
- Headed the Debugging Task Force; trained new hires on internal debugging tools and revised the onboarding curriculum

Intern – Willamette University, Physics Department May 2018 – July 2018

- Implemented 3D wavelet convolutions via FFT in Python to search for baryon acoustic oscillation (BAO) signals in the 2MASS Redshift Survey – the first application of this method to the 2MRS
- Applied Gaussian smoothing, zero-padding, and a custom local maxima detection algorithm to identify candidate BAO locations from a cosmological density field
- Built a computer for parallel GPU processing to support analysis of a large-scale cosmological dataset

TECHNICAL SKILLS

- **Languages**: Python, C#, Java, HTML, CSS, JavaScript, SQL, M/MUMPS
- **AI & ML:** NumPy, Matplotlib, pandas, Stable-Baselines3, Gymnasium, PyTorch, scikit-learn, OpenAI API, Anthropic API, Prompt Engineering, LangGraph, Chroma, RAG
- **Frameworks & Tools:** Django, FastAPI, Git, Azure, Linux, nginx, GitHub Actions (CI/CD), LaTeX
- **Healthcare:** Epic, revenue cycle, X12 EDI (837I/837P, 835), HIPAA/PHI, FHIR, Medicare coverage policies (LCD/NCD)

CERTIFICATIONS

- **Microsoft Certified: Azure Fundamentals (AZ-900)** October 2026

PROJECTS AND RESEARCH

- **Predicting Medicare Coverage from FHIR Records with RAG and LLMs**
  Healthcare AI Research, 2026
  - Built a prior authorization checker that retrieves governing Medicare coverage policies using RAG, judging each clinical criterion in a FHIR record as met, not met, or insufficient information

- Reached 98-99.5% policy selection and 89-94% decision accuracy on 72 synthetic cases (development and held-out policies), with under 3% false "covered" decisions
- Deployed to an Azure VM (nginx, gunicorn, systemd, HTTPS) with CI/CD via GitHub Actions and SQLite request logging
- **Tools**: Python, FastAPI, Chroma, Anthropic API, FHIR, Azure, GitHub Actions, SQLite

- **Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems**
  Educational Technology Research, 2025-2026
  - Built a web application generating personalized coding exercises to improve code comprehension
  - Designed and implemented a multi-model pipeline with conditional routing, prompt chaining, guardrails, and feedback loops across OpenAI and Anthropic APIs
  - Built a validation and repair loop that executes generated code, verifies it against user specifications via AST inspection, and escalates to a stronger model on failure
  - **Tools**: Python, Django, JavaScript, HTML/CSS, OpenAI and Anthropic APIs

- **Improving Users' Academic Writing Ability Through LLM-Generated Feedback**
  Educational Technology Research, 2026
  - Built a Django web application that generates writing feedback by decomposing analysis across six specialized LLM agents using LangGraph, with a synthesis step reconciling critiques
  - Designed a controlled evaluation (position-bias and cross-family-judge controls) that isolated analyzer strength as the performance bottleneck
  - Implemented RAG with Chroma (genre exemplars, rubric criteria) and tested it via ablation; found no net benefit at n=10 and reported it as a null result
  - **Tools**: Python, Django, LangGraph, Chroma, HTML/CSS, OpenAI and Anthropic APIs

- **Benchmarking Reinforcement and Imitation Learning on a Deterministic Platformer: Jump King**
  Reinforcement Learning Research, 2026
  - Built a full pipeline interfacing with a live commercial game through real-time state extraction (custom C# mod), TCP socket communication, and a custom Gymnasium environment
  - Designed and implemented behavioral cloning (BC) from human playthroughs, per-level reward shaping, and a class-based framework for training and evaluating RL, BC, and BC+RL agents
  - Chained per-level agents reliably complete the full 43-level base game in ~10 minutes
  - Showed BC initialization solves levels where pure RL fails entirely, and converges far faster elsewhere
  - **Tools**: Python, C#, Stable-Baselines3, Gymnasium, dnSpy, .NET

POSTERS AND PRESENTATIONS

- *Improving Students' Code Comprehension Ability Through LLM-Generated Practice Problems*
  Poster Presentation – National Council on Measurement in Education (April 2026)
- *Looking for the BAO Signal in the 2MRS Using the Wavelet Transformation*
  Oral Presentation – Willamette-Linfield intercollegiate symposium (May 2019)
  Poster Presentations – 223rd Meeting of the American Astronomical Society (January 2019), Murdock College Science Research Conference (November 2018)